

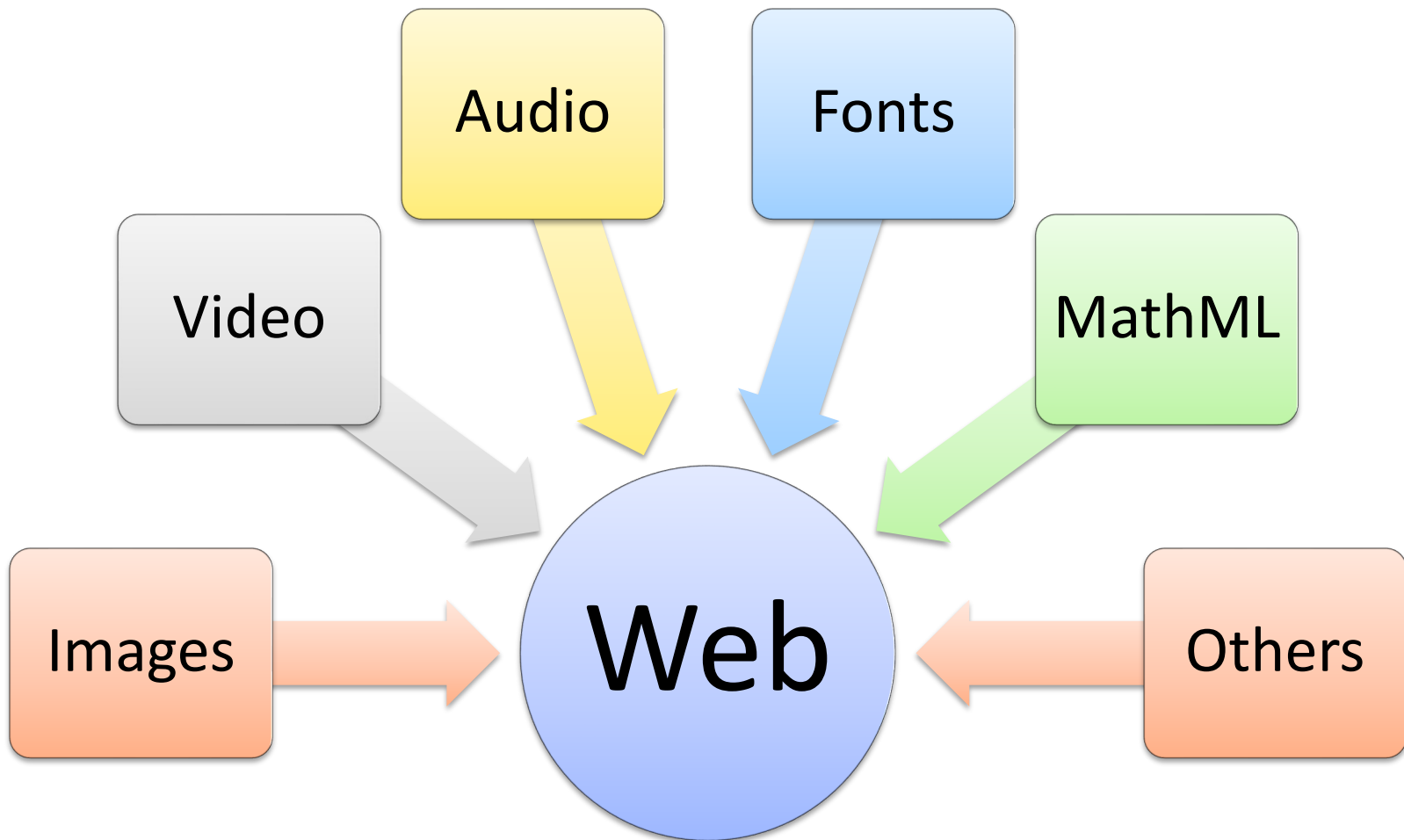


WIESŁAW PIETRUSZKIEWICZ

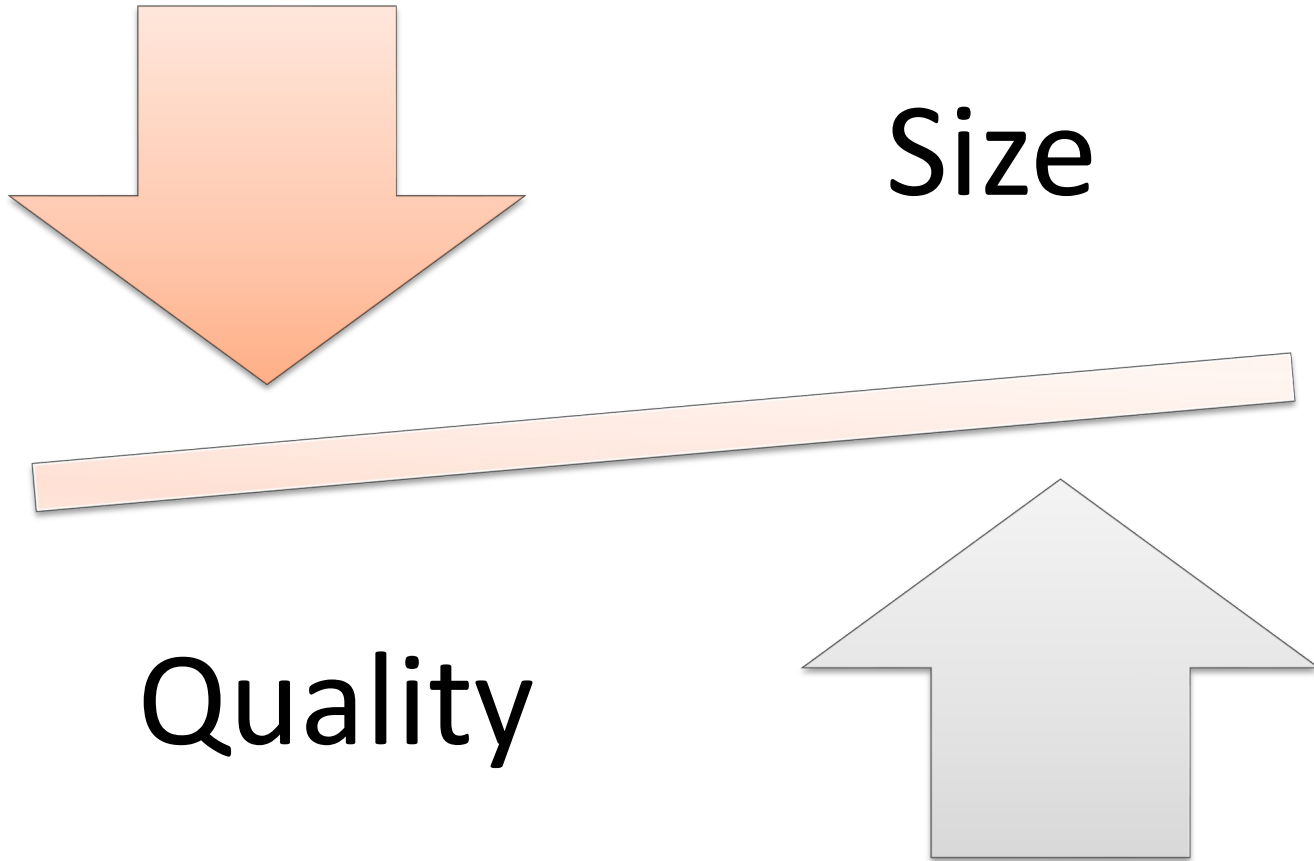
Role of media elements on the Web



Media elements



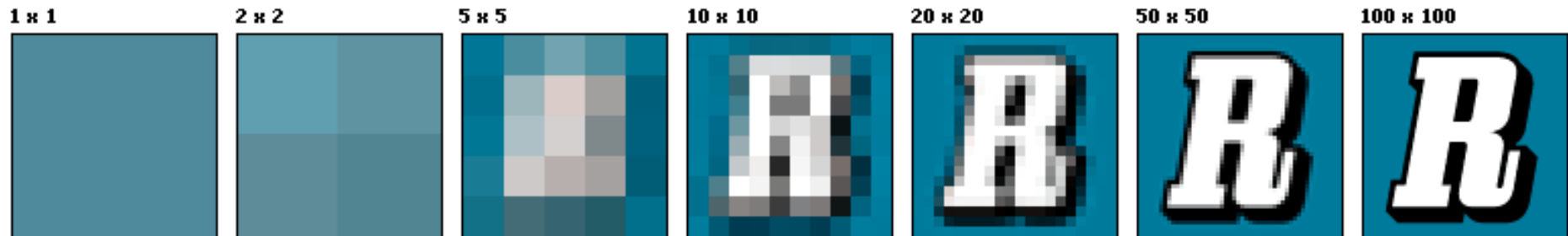
Compromise



WEB IMAGES

Raster images – Basic parameters

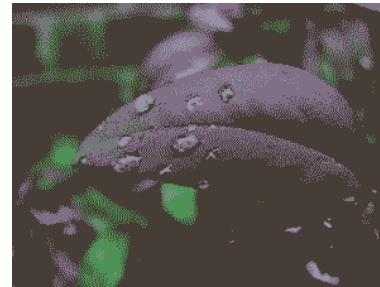
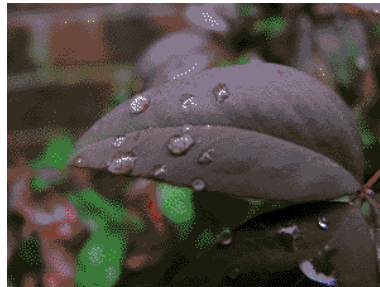
- Resolution is a basic parameter defining number of pixels
- Pixels contain information about details
- More pixels = more details = larger image size



Source: wikipedia.org

Raster images – Basic parameters

- Colour Depth indicate number of bits used to represent colour of single pixel
- More colours = larger image size

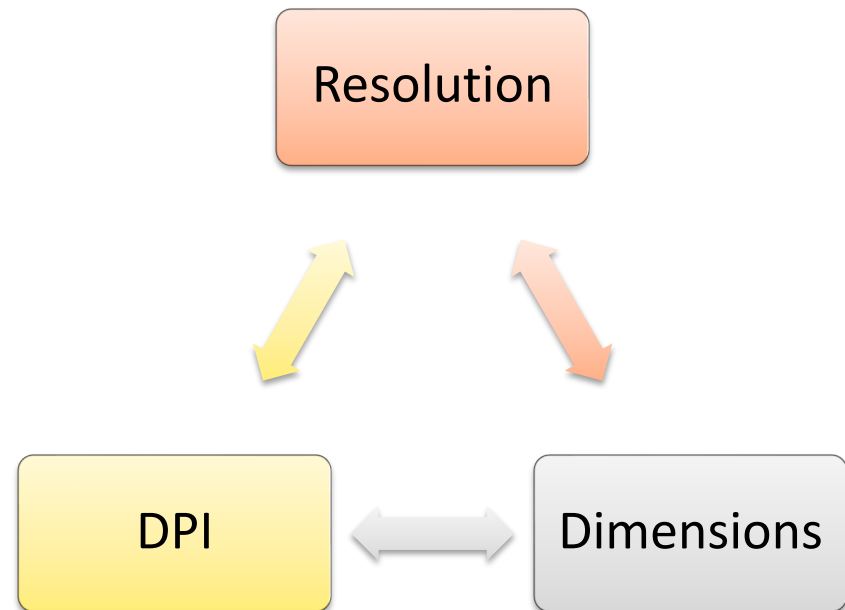


Source: wikipedia.org

Raster images – Basic parameters

Dots Per Inch (DPI)

- Typical Web 72 dpi
- Printed materials 200-300 dpi
- An optimal solution is when all 3 match



Compression

- Lossless compression allows perfect reconstruction of original data
- Used to compress text, programmes
- Used to compress multimedia (if desired)
- Usually larger compressed multimedia
- Lossy compression allows approximated reconstruction of original data
- Used in multimedia
- Based on limited human perception
- Quality vs size

Graphics Interchange Format - GIF

- 8 bits per pixel (indexed 256 colour from 24-bit RGB colour space)
- Lossless compression (LZW)
- Animation by frames
- Preliminary web standard for logos/schemas

PNG

- Successor of GIF – non-patented replacement
- Lossless compression
- 24-bit RGB, 32-bit RGBA(Alpha channel) or greyscale images
- Standard offers transparency
- It is possible to store animations (APNG)

JPEG

- Commonly used lossy compression for images
 - Lossless JPEG (has not become popular)
- 8-bit (per channel) file = 24 bits depth
- Q (quality) factor

				
Q = 100 81447 bytes 2.7:1	Q = 50 14679 bytes 15:1	Q = 25 9407 bytes 23:1	Q = 10 4787 bytes 46:1	Q = 1 1523 bytes 144:1

WebP

- Lossy or lossless compression
- Developed by Google
- Universal replacement for GIF, PNG & JPEG
- Possible to use animation
- Alpha transparency available

WEB VIDEO & AUDIO

A structure of movie (frames)

- We have 2 types of video coding:
 - Intraframe coding – where each frame is independent of the previous one → larger size, flexible editing
 - Interframe coding – where the differences between frames is encoded (delta frames), and periodic information about entire image is added (keyframes) → smaller size, limited editing

Web Video Containers

- Container = file structure storing video content and metacontent
- There is many containers around – some partially supported by different browsers
- Popular containers with wide support:
 - MPEG-4
 - Ogg
 - WebM
 - MKV (Open Source, multiple audio layers, flexible, ~~HTML5~~)

Video Codecs for Web

Codec name (short)	Full codec name	Container support
AV1	AOMedia Video 1	MP4 , WebM
AVC (H.264)	Advanced Video Coding	3GP , MP4
H.263	H.263 Video	3GP
HEVC (H.265)	High Efficiency Video Coding	MP4
MP4V-ES	MPEG-4 Video Elemental Stream	3GP , MP4
MPEG-1	MPEG-1 Part 2 Visual	MPEG , QuickTime
MPEG-2	MPEG-2 Part 2 Visual	MP4 , MPEG , QuickTime
Theora	Theora	Ogg
VP8	Video Processor 8	3GP , Ogg , WebM
VP9	Video Processor 9	MP4 , Ogg , WebM

Source: developer.mozilla.org

Constant vs Variable Bitrate

- Constant bitrate defines a fixed number of bits to encode multimedia content
- Stable size in time
- Easier to apply
- Larger files
- Inefficient storage
- Wider support
- Variable bitrate uses a variable number of bits depending on content
- Stable quality in time
- More complex
- Smaller files
- Efficient storage
- Incompatibility issues

Factors Affecting Multimedia

Resolution

- Effects the quality
- Larger resolution - more information to store

Colour depth

- Effects the quality
- Deeper depth - more information to store

Motion

- Depends on content
- More dynamics - more information is necessary

Key frames

- Faster selection of frames
- More frequent key frames - more information to store

Frame rate

- Effects the smoothness of movie
- More frames - more information to store

Noise

- Effects the quality
- Makes compression harder

Dynamic Documents and Front-end Web Development

Multimedia – Common resolutions (Handbrake)

Preset	Type	Video	Audio	Picture Quality	Encoding Speed	File Size
Discord Nitro Large 3-6 Minutes 1080p30	MP4	H.264	AAC stereo	Depends on source	Fast	50/100 MB or less
Discord Nitro Medium 5-10 Minutes 720p30	MP4	H.264	AAC stereo	Depends on source	Fast	50/100 MB or less
Discord Nitro Small 10-20 Minutes 480p30	MP4	H.264	AAC stereo	Depends on source	Fast	50/100 MB or less
Discord Small 2 Minutes 360p30	MP4	H.264	AAC mono	Depends on source	Very Fast	8 MB or less
Discord Tiny 5 Minutes 240p30	MP4	H.264	AAC mono	Depends on source	Very Fast	8 MB or less
Gmail Large 3 Minutes 720p30	MP4	H.264	AAC stereo	Depends on source	Fast	25 MB or less
Gmail Medium 5 Minutes 480p30	MP4	H.264	AAC stereo	Depends on source	Fast	25 MB or less
Gmail Small 10 Minutes 288p30	MP4	H.264	AAC mono	Depends on source	Fast	25 MB or less
Vimeo YouTube HQ 2160p60 4K	MP4	H.264	AAC stereo	High	Medium	Large
Vimeo YouTube HQ 1440p60 2.5K	MP4	H.264	AAC stereo	High	Medium	Large
Vimeo YouTube HQ 1080p60	MP4	H.264	AAC stereo	High	Medium	Large
Vimeo YouTube HQ 720p60	MP4	H.264	AAC stereo	High	Medium	Large
Vimeo YouTube 720p30	MP4	H.264	AAC stereo	Standard	Medium	Average

Web Media & Fonts

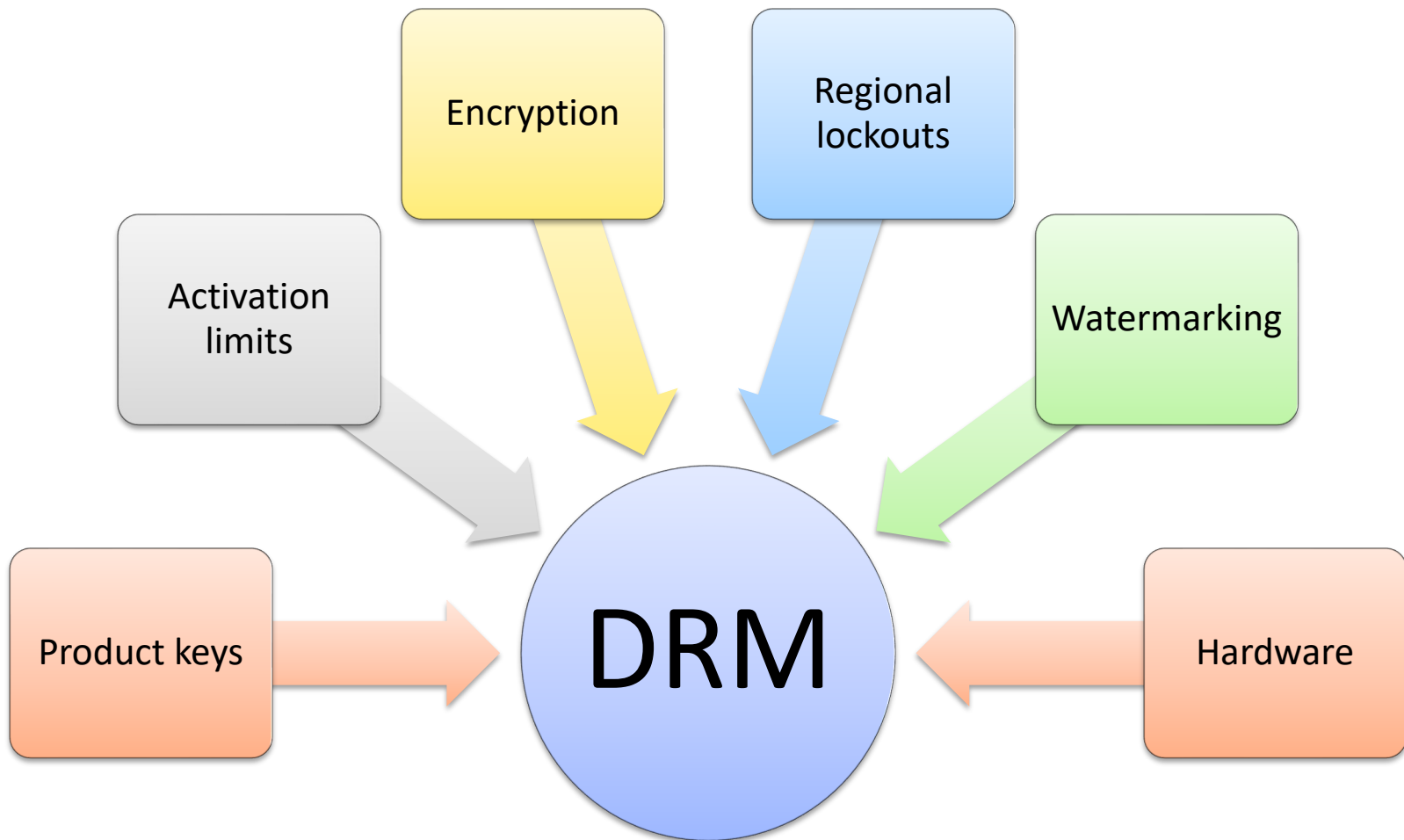
More Information about Compression

- For more information about compression:
https://developer.mozilla.org/en-US/docs/Web/Media/Formats/Video_codecs

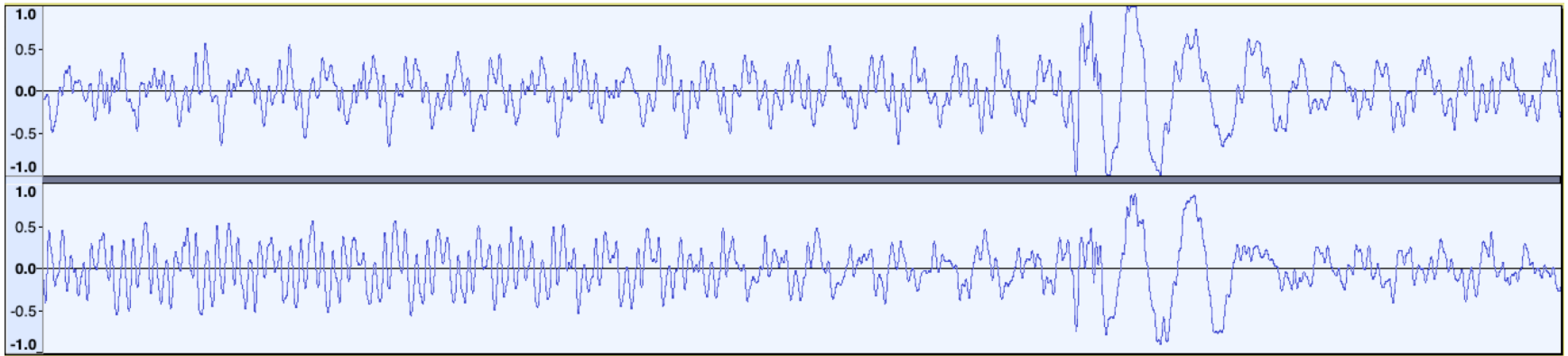
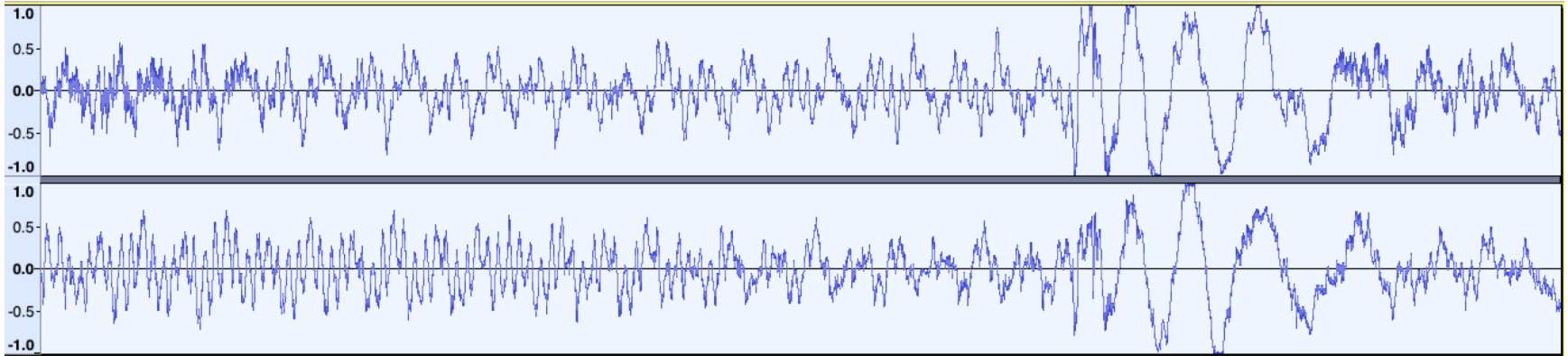
Digital Rights Management (DRM)

- DRM is a set of measures allowing copyright holders to manage the access to digital content
- DRM is supposed to help prevent copyright infringement
- Main idea – every situation, when content is to be watched must be approved by copyright holders

DRM – Spectrum of technologies



Audio (sample with a bandpass filter)



Audio for Web

- Popular multimedia containers allow to store audio:
 - MPEG-4
 - OGG
 - WebM
- Popular MP3 audio files are MPEG-4s with MPEG-1 Audio Layer III
- AAC is recommended
- Factors influencing quality and size of audio:
 - Channel count
 - Noise/hiss
 - Sample rate
 - Sample size (bits)

Audio – Quality examples

- There are common suggestions relating to the usage of audio for various purposes:
 - Talk 32(AAC)/64 (MP3) kbps
 - Podcasts 64 (AAC) / 96 (MP3) kbps
 - Standard music 96 (AAC) / 128 (MP3) kbps
 - HQ music up to 240 (AAC) / 320 (MP3) kbps
- Frequency range of speakers/headsets limits
- Human range 20–20 kHz (changes in time)

WebRTC

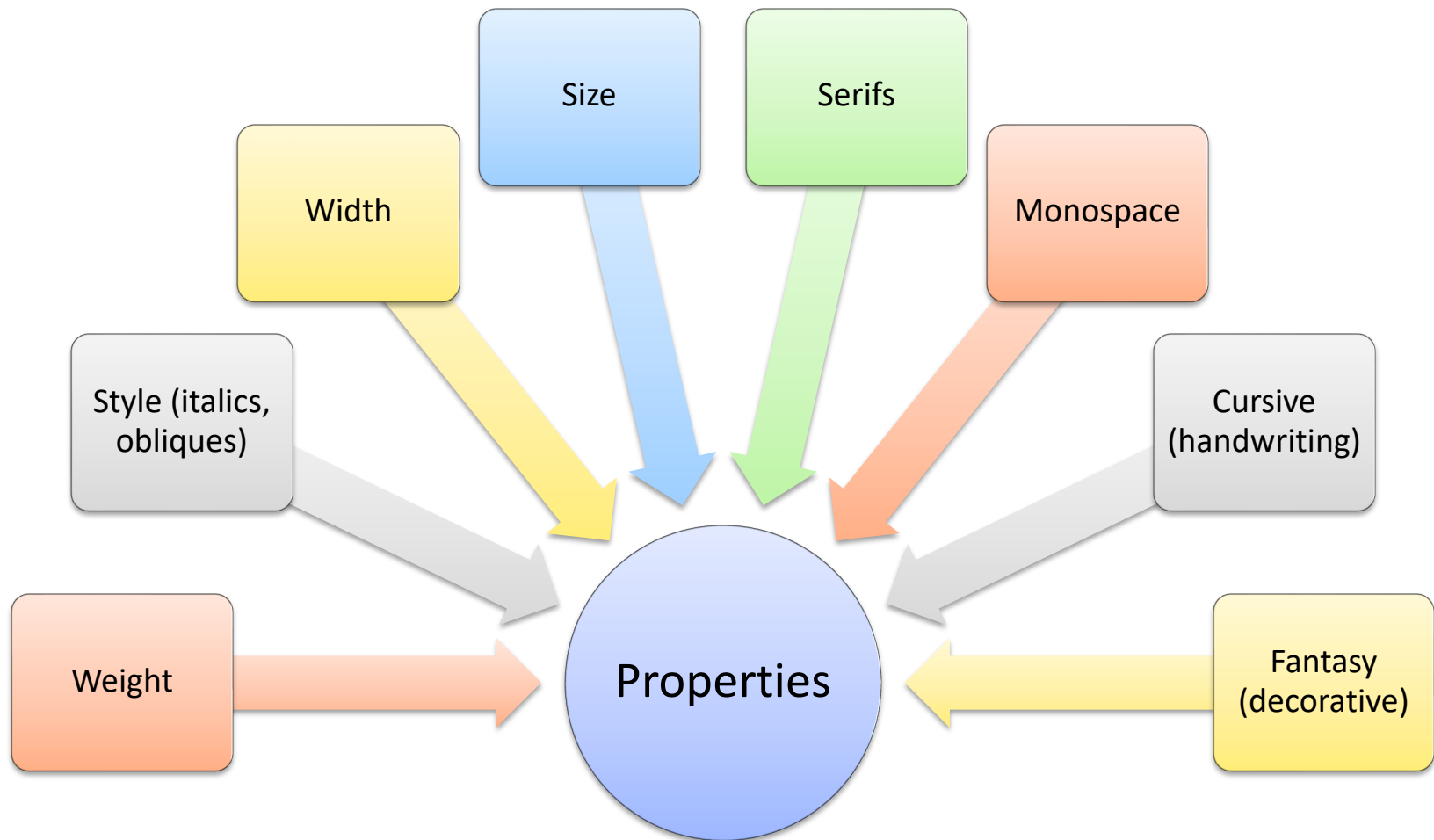
- Web Real-Time Communication is a standard to connect Peer-To-Peer nodes and allow video streaming via Web browser
- A part of HTML5 standard
- Controlled by JavaScript API
- Supported video codecs: H.264, VP8 & VP9
- Supported audio codes: Opus, G.711 PCM (A-law), G.711 PCM (μ -law), G.722, iLBC, iSAC

WebRTC usage

- Common for web-based video chats platforms
– Zoom, MS Teams, Slack etc.
- Supported by the majority of currently used web browsers
- No need to have dedicated applications
- No need to use additional plugins (codecs)

OTHER VISUAL ELEMENTS

Fonts – Common features



Fonts for Web Pages

- If specific fonts are used by web page, they may not be available on client's browser
- Hence, the fonts must be distributed with web content
- WOFF – embedded TrueType or OpenType compressed fonts
- EOT – Embedded OpenType
- TTF – TrueType
- OTF – OpenType
- SVG – Scalable Vector Graphics
- WOFF – Web Open Font Format (W3C standard)

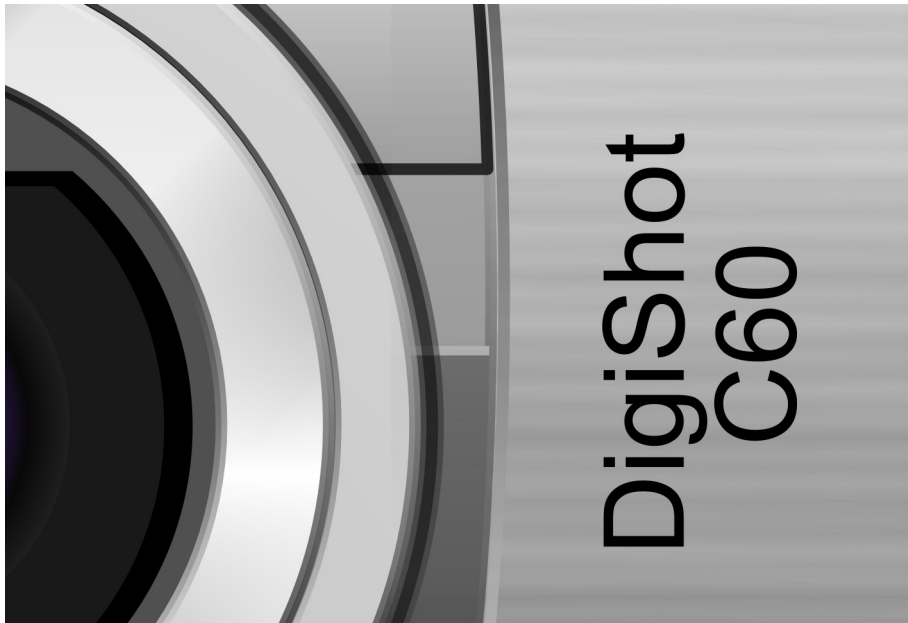
Web safe fonts (the core list)

- There are various lists of web safe fonts
- The core items are:
 - Sans serif fonts: Arial, Tahoma, Trebuchet MS, Verdana
 - Serif fonts: Georgia, Times New Roman
 - Monospaced font: Courier New
- IF YOU HAVE TO USE ANY OTHER FONTS — *distribute it with your page*

SVG - Standard

- Scalable Vector Graphics – an XML-based vector image format
- Support for interactivity and animation
- Scalable content
- Possible compression – SVGZ
- Pre-HTML5 way to draw on client-side
- Functionally replaced by Canvas proposed by HTML5

SVG vs Raster - Example



SVG vs Canvas

SVG

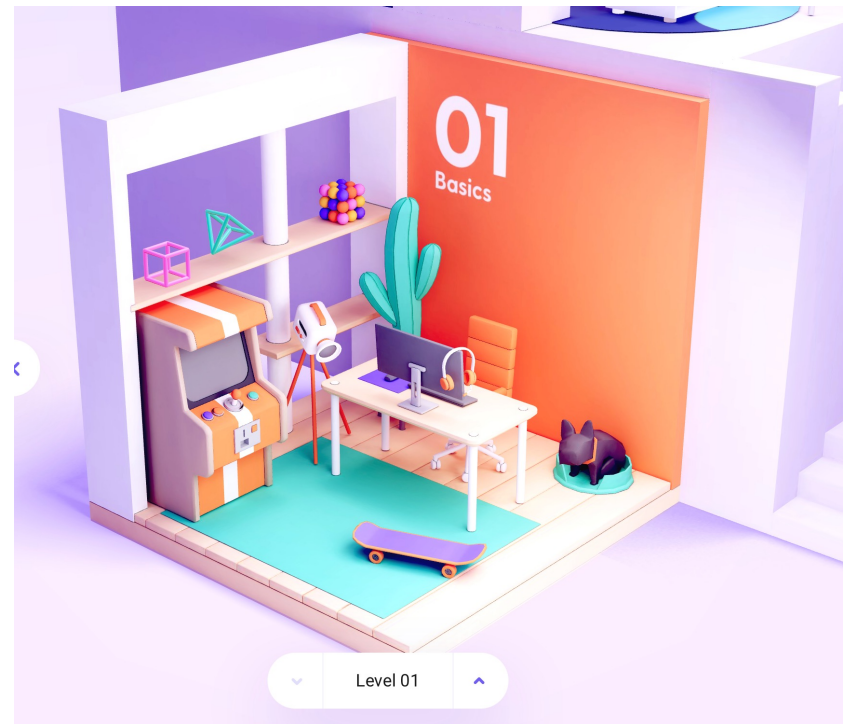
- Used as a vector graphics format
- A graphic-oriented approach
- Drawing done by creation of XML content by JS libraries
- Mostly static
- Supported by various tools (including non-web ones)

Canvas

- Drawing element
- A programming-oriented approach
- Actual drawing done by JavaScript
- Possible animations
- Stays at the Web environment

WebGL

- Web Graphics Library is a JavaScript API for creation and rendering 2D or 3D graphics
- Development support offered by frameworks:
 - Three.js
 - Unity
 - Babylon.js
 - & others

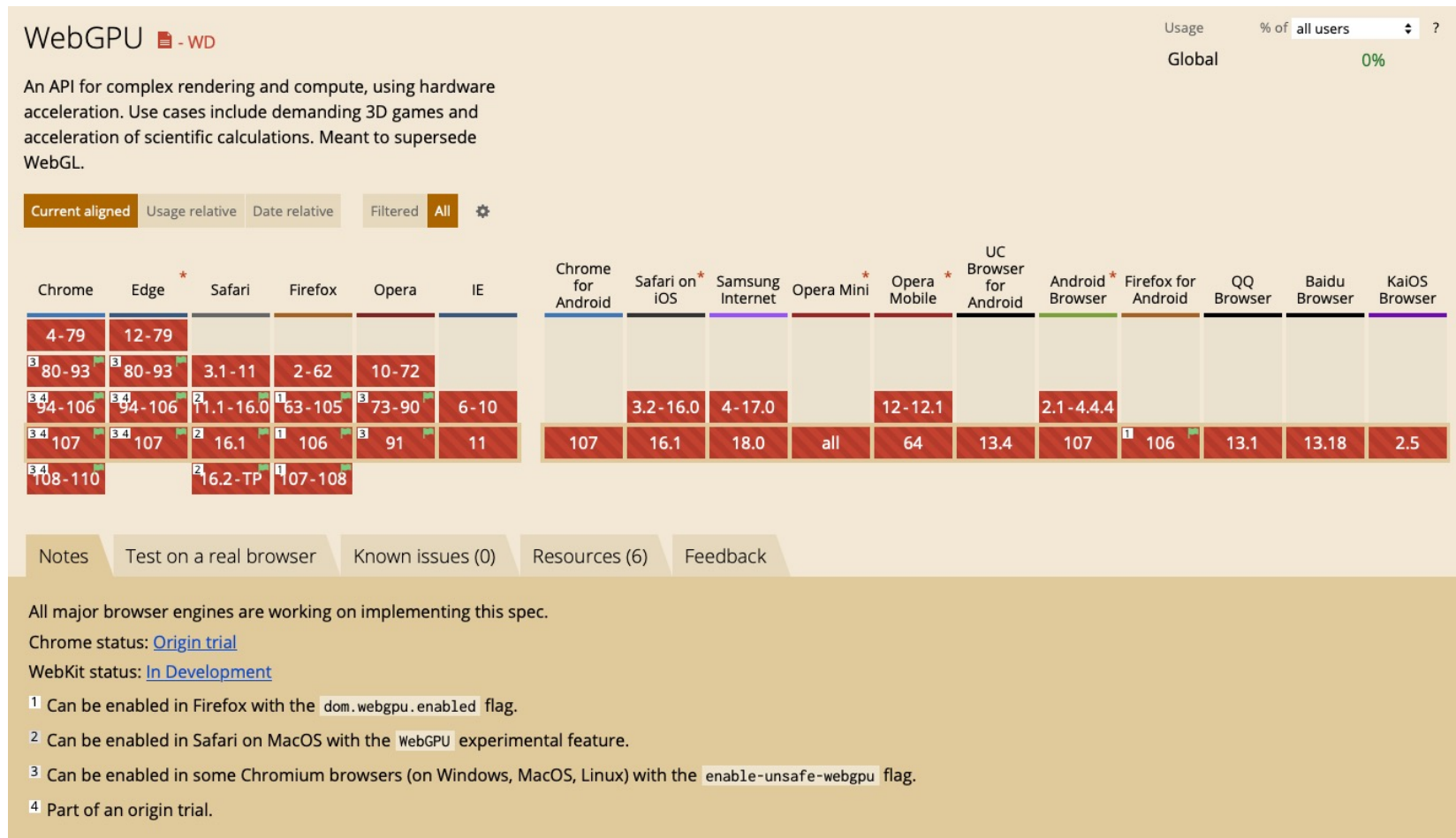


An example of WegGL (three.js-based):
threejs-journey.com

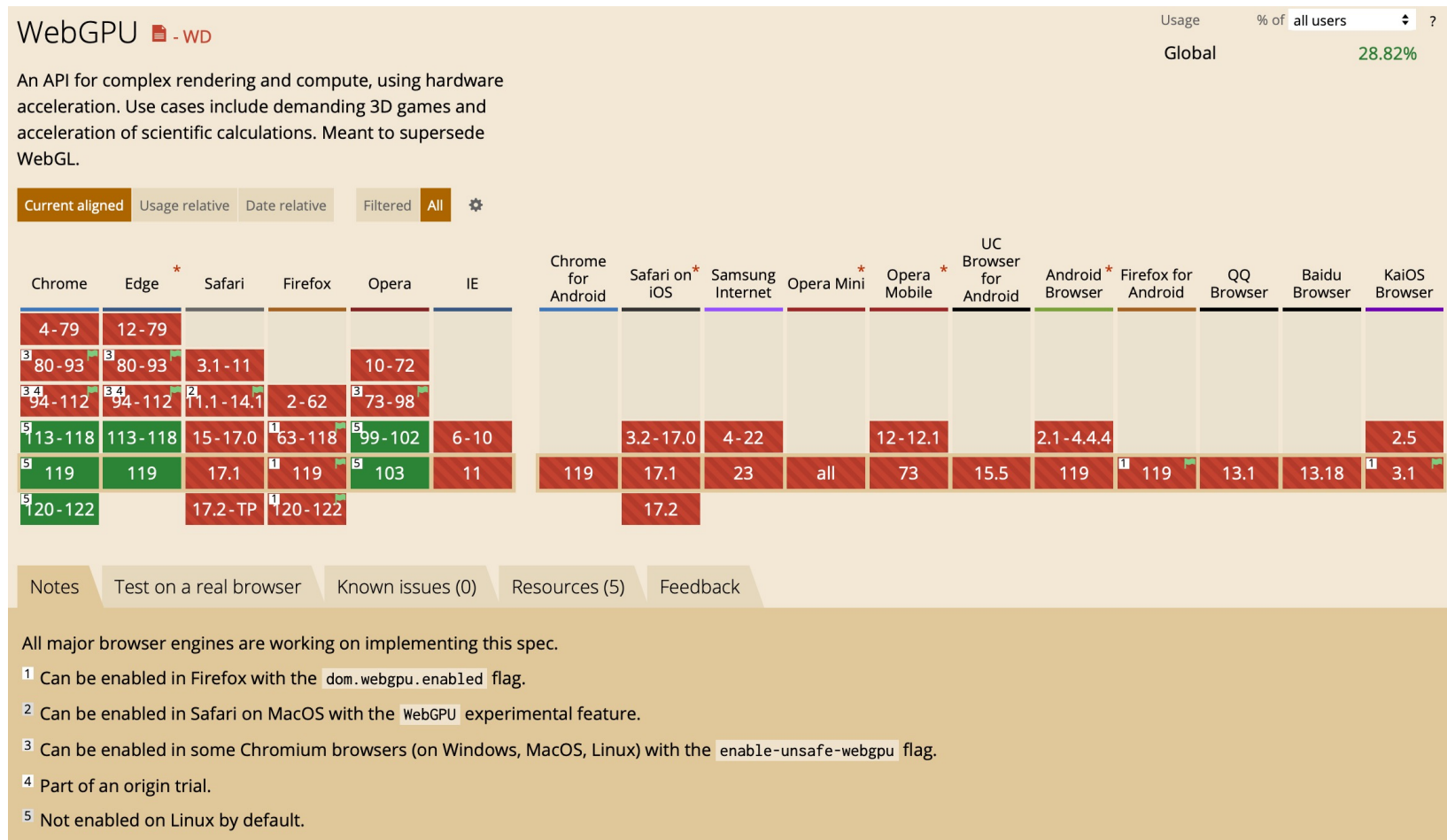
WebGPU

- Idea – use GPU resources by browser
- Started in 2021 – currently developed successor of WebGL
- Still a working draft in 2022
- Not limited to visualisation, as it includes possibility of calculations
- A huge potential – it will take some time to prepare the other pieces (browsers, libraries)

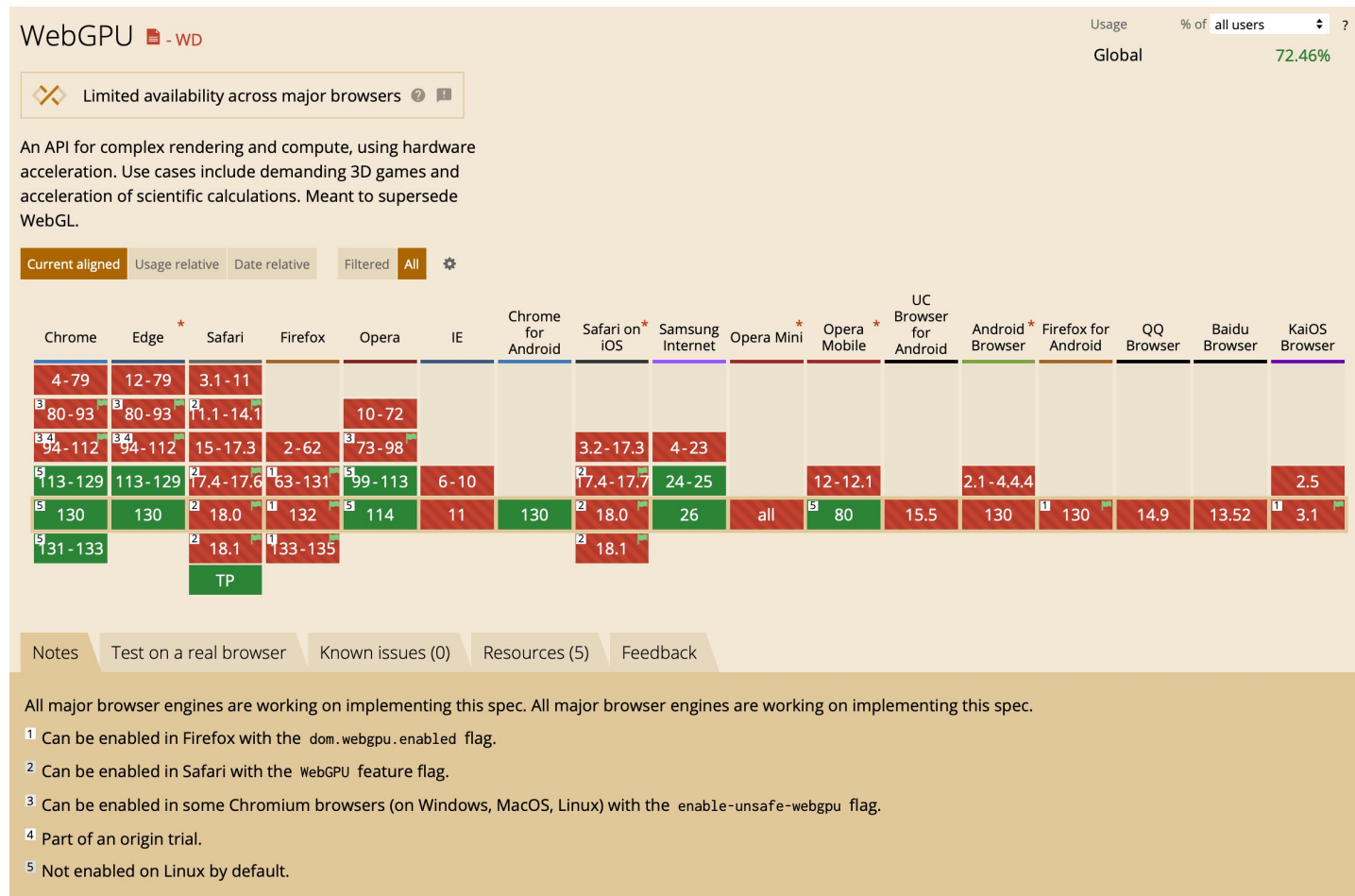
Support for WebGPU in 2022



Support for WebGPU in 2023



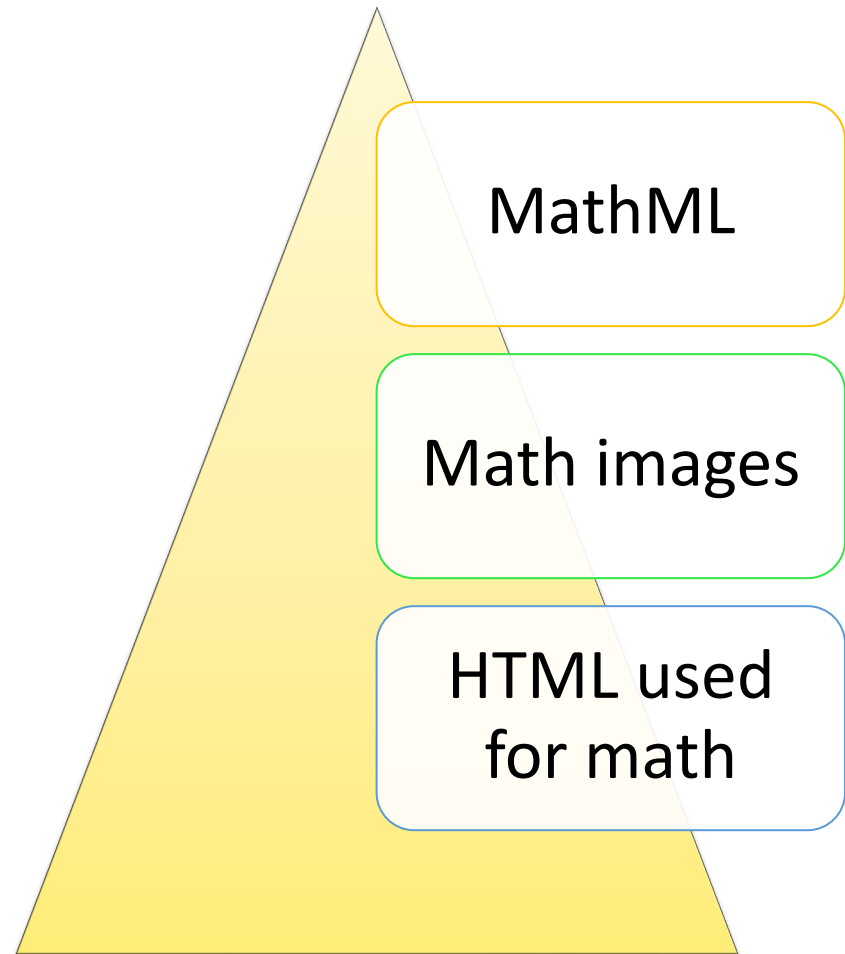
Support for WebGPU in 2024



MATHML

MathML – Idea

- Mathematical Markup Language (MathML) is a mathematical markup language used to describe mathematical notations
- It is an application of XML
- Main usage – the Web



MathML – Usage

- The presentation & content are separated
- This is a way to provide information about math expressions, allowing its presentation to be done in different ways (displayed/read)
- An essential presentation element – rule of thumb: use Latin Modern Math and STIX Two Math (both prepared to be used for math on Web pages)

MathML – Examples (w3c.org)

Presentation MathML

```
<mrow>
  <mrow>
    <msup>
      <mi>x</mi>
      <mn>2</mn>
    </msup>
    <mo>+</mo>
    <mrow>
      <mn>4</mn>
      <mo>&InvisibleTimes;</mo>
      <mi>x</mi>
    </mrow>
    <mo>+</mo>
    <mn>4</mn>
  </mrow>
  <mo>=</mo>
  <mn>0</mn>
</mrow>
```

$$x^2 + 4x + 4 = 0$$

Content MathML

```
<mrow>
  <apply>
    <eq/>
    <apply>
      <plus/>
      <apply>
        <power/>
        <ci>x</ci>
        <cn>2</cn>
      </apply>
      <apply>
        <times/>
        <cn>4</cn>
        <ci>x</ci>
      </apply>
      <cn>4</cn>
    </apply>
    <cn>0</cn>
  </apply>
</mrow>
```


MathML – Miscellaneous

- Manual preparation of MathML is problematic (for more complex math expressions)
- Consider it to be a language used to present math, not something to be written directly
- Use additional tools
- Or convert from other notation languages (mainly LaTeX)

MathML – Authoring example

The screenshot displays the MyScript MathML authoring interface. At the top, a navigation bar includes a back arrow, a "Math" tab with a calculator icon, and an information icon. The main workspace is divided into three sections. The top section shows the typed equation $a^2 + b^2 = c^2$. The middle section contains a blue "Convert" button. The bottom section shows the handwritten equation $a^2 + b^2 = c^2$. To the right, a panel with tabs for "Math", "LaTeX", and "MathML" displays the corresponding MathML code:

```
<math
xmlns='http://www.w3.org/1998/Math/MathML'>
<msup> <mrow> <mi> a </mi> </mrow> <mrow>
<mn> 2 </mn> </mrow> </msup> <mo> + </mo>
<msup> <mrow> <mi> b </mi> </mrow> <mrow>
<mn> 2 </mn> </mrow> </msup> <mo> = </mo>
<msup> <mrow> <mi> c </mi> </mrow> <mrow>
<mn> 2 </mn> </mrow> </msup> </math>
```

Below the code is a grid area with a blue copy icon. The bottom of the interface features the MyScript logo, copyright information, and a "POWERED BY desmos" logo.

TEMML – LaTeX to MathML

TE^MML

A TeX-to-MathML conversion library in JavaScript.

Input:

```
\def\d{\mathrm{d}}

\oint_C {\vec{B}\circ\vec{l}} =
\mu_0 \left( I_{\text{enc}} +
\varepsilon_0 \frac{d}{dt} \int_S
{\vec{E}\circ\hat{n}}; da
\right)
```



☐ Wide

Output:

$$\oint_C \vec{B} \circ d\vec{l} = \mu_0 \left(I_{\text{enc}} + \varepsilon_0 \frac{d}{dt} \int_S \vec{E} \circ \hat{n} da \right)$$

☒ Display Mode

Latin Modern

Math



☐ Annotate

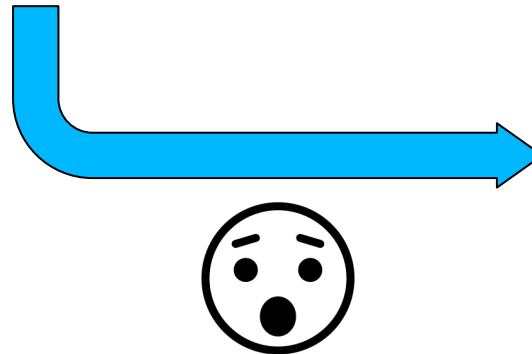
☐ XML

- Fast
- Lightweight
- Extensive LaTeX coverage
- Thoroughly tested
- Works in browser or server-side with Node.js
- No dependencies
- Use any math font
- MathML is now supported by Chrome, Edge, Firefox, and Safari.
(Safari has some rendering bugs.)

TEMLML - Example

```
\def\d{\mathrm{d}}
```

```
\oint_C {\vec{B}}\circ \d\vec{l} = \mu_0  
\left( I_{\text{enc}} + \varepsilon_0  
\frac{\d}{\d t} \int_S {\vec{E}} \circ  
\hat{n} \right) \d a \right)
```



```
<math display="block" class="tml-display" style="display:block math;">  
<mrow>  
<msub>  
<mo movablelimits="false"></mo>  
<mi>C</mi>  
</msub>  
<mrow>  
<mover>  
<mi>B</mi>  
<mo stretchy="false" style="transform:scale(0.75) translate(10%, 30%);">~</mo>  
</mover>  
<mo>~</mo>  
</mrow>  
<mrow>  
<mi mathvariant="normal">d</mi>  
</mrow>  
<mover>  
<mi>l</mi>  
<mo stretchy="false" style="transform:scale(0.75) translate(10%, 30%);">~</mo>  
</mover>  
<mo>~</mo>  
<msub>  
<mi>u</mi>  
<mo>0</mn>  
</msub>  
<mrow>  
<mo fence="true" form="prefix"></mo>  
<msub>  
<mi>I</mi>  
<mtext>enc</mtext>  
</msub>  
<mo>+</mo>  
<msub>  
<mi>e</mi>  
<mo>0</mn>  
</msub>  
<mfrac>  
<mi>~</mi>  
</mfrac>  
<msub>  
<mo movablelimits="false"></mo>  
<mi>S</mi>  
</msub>  
<mrow>  
<mi mathvariant="normal">d</mi>  
</mrow>  
<mrow>  
<mi>E</mi>  
<mo stretchy="false" style="transform:scale(0.75) translate(10%, 30%);">~</mo>  
</mover>  
<mo>~</mo>  
<mover>  
<mi>n</mi>  
<mo stretchy="false" class="tml-xshift" style="math-style:normal;math-depth:0;">^</mo>  
</mover>  
</mrow>  
<mspace width="0.2778em"></mspace>  
<mrow>  
<mi mathvariant="normal">d</mi>  
</mrow>  
<mi>a</mi>  
<mo fence="true" form="postfix"></mo>  
</mrow>  
</math>
```

TeXZilla

TeXZilla

Description

[TeXZilla](#) is a Javascript LaTeX-to-MathML converter which is compatible with Unicode. It is generated by [Jison](#) using a grammar similar to the one of [itex2MML](#) and the [XML Entity Definitions for Characters](#) specification.

Live Demo

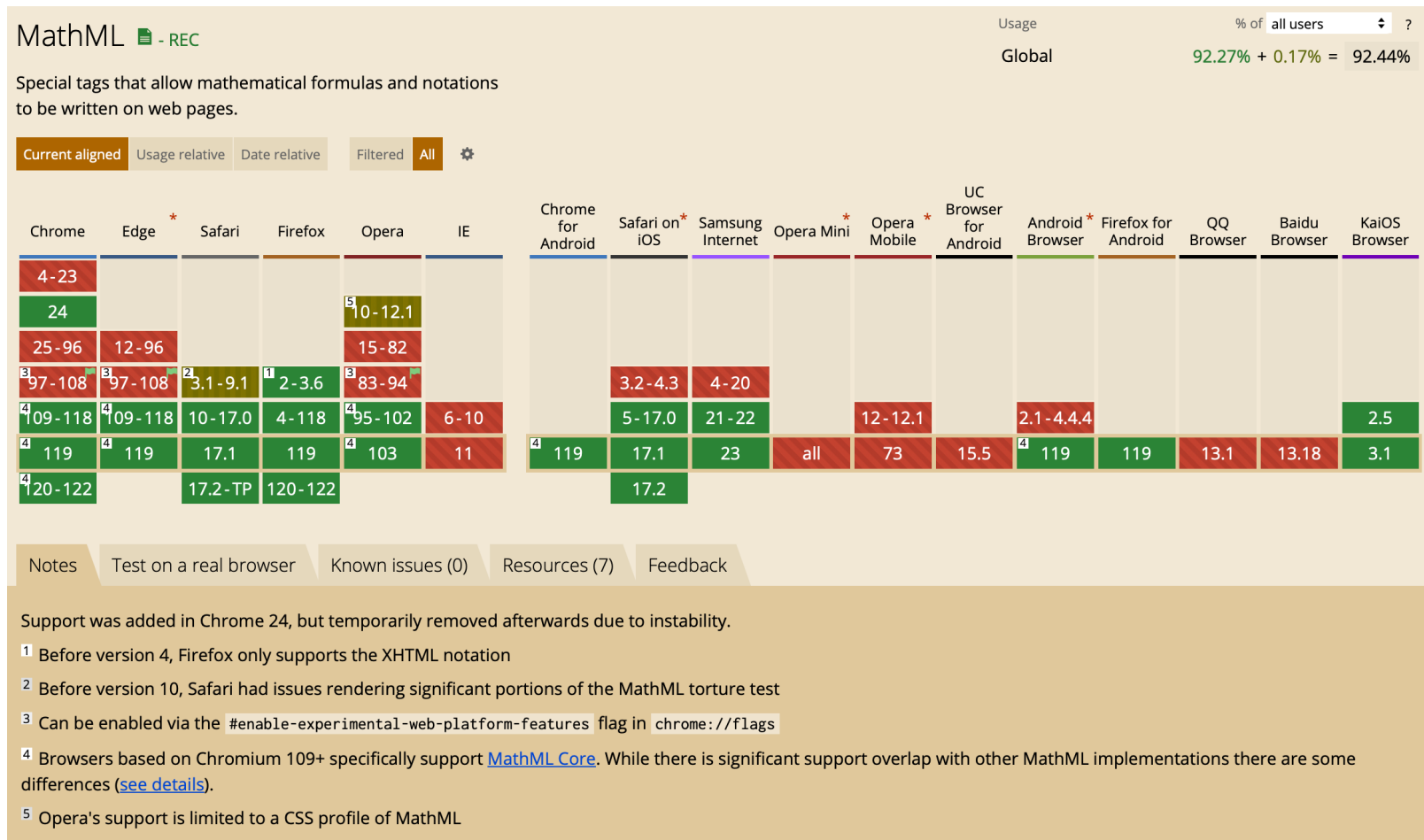
Enter your LaTeX source or pick an example and check how the MathML output is rendered by your browser.

Display Mode ; LTR ;
Font: (see [Fonts for Mozilla's MathML engine](#))

$f(x)=\sum_{n=-\infty}^{\infty} c_n e^{2\pi i (n/T) x} = \sum_{n=-\infty}^{\infty} \hat{f}(\xi_n) e^{2\pi i \xi_n x} \Delta \xi$

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{2\pi i (n/T) x} = \sum_{n=-\infty}^{\infty} \hat{f}(\xi_n) e^{2\pi i \xi_n x} \Delta \xi$$

Support for MathML

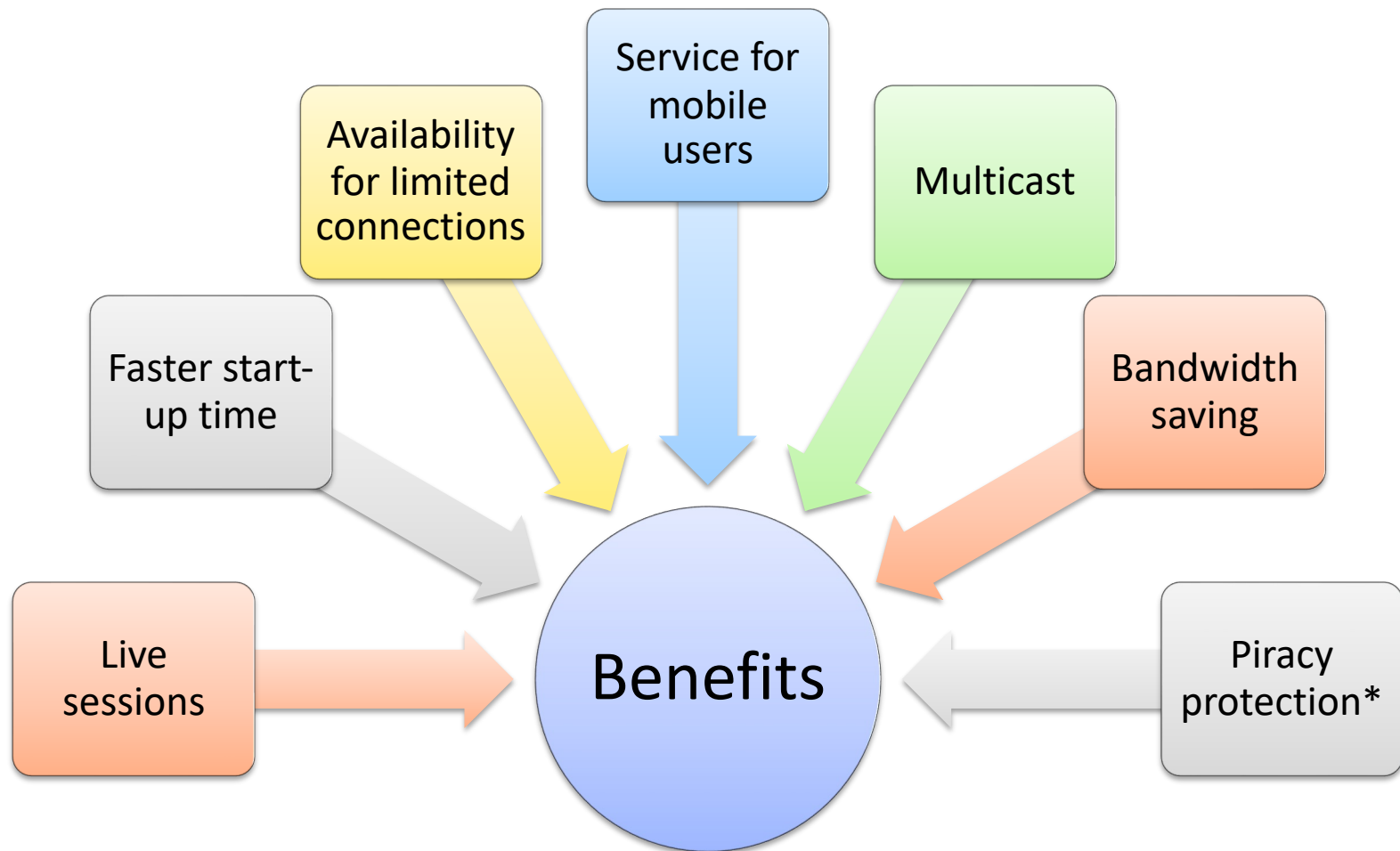


STREAMING

Streaming – Ideas

- On demand – deliver content partially and present it not waiting for the rest to be downloaded
- Live streaming – deliver content as it is created
- A buffer protects against unstable connection
- Different streams - various versions of multimedia (resolution & bit rate)

Streaming – Benefits



Streaming – Example

The screenshot shows a web browser window with the YouTube video player. The video title is "EARTH in 8K ULTRA HD - Tour Through the Planet Earth - Best Places and Animals Relaxing Music 8K TV". The video is playing at 0:19 / 30:18. A quality menu is open, showing options from 240p to 4320p 8K. The video has 5,188,839 views and was uploaded on May 7, 2021. The channel is "8K VIDEOS ULTRA HD".

Quality menu options:

- 4320p 8K
- 2160p 4K
- 1440p HD
- 1080p HD
- 720p
- 480p
- 360p
- 240p

Video description:

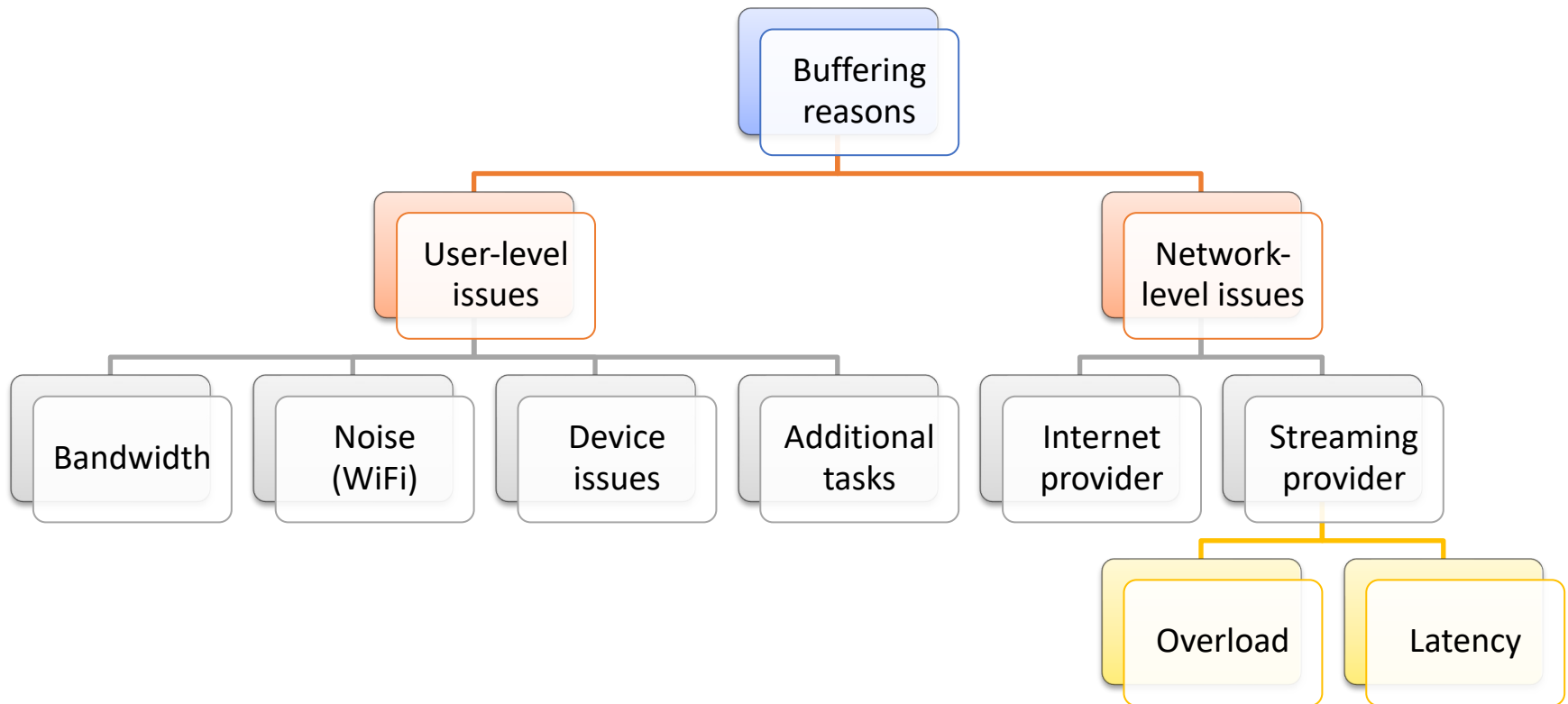
#8K #PLANET #EARTH
EARTH in 8K ULTRA HD - Tour Through the Planet Earth - Best Places and Animals Relaxing Music 8K TV
5,188,839 views • May 7, 2021

Channel: 8K VIDEOS ULTRA HD

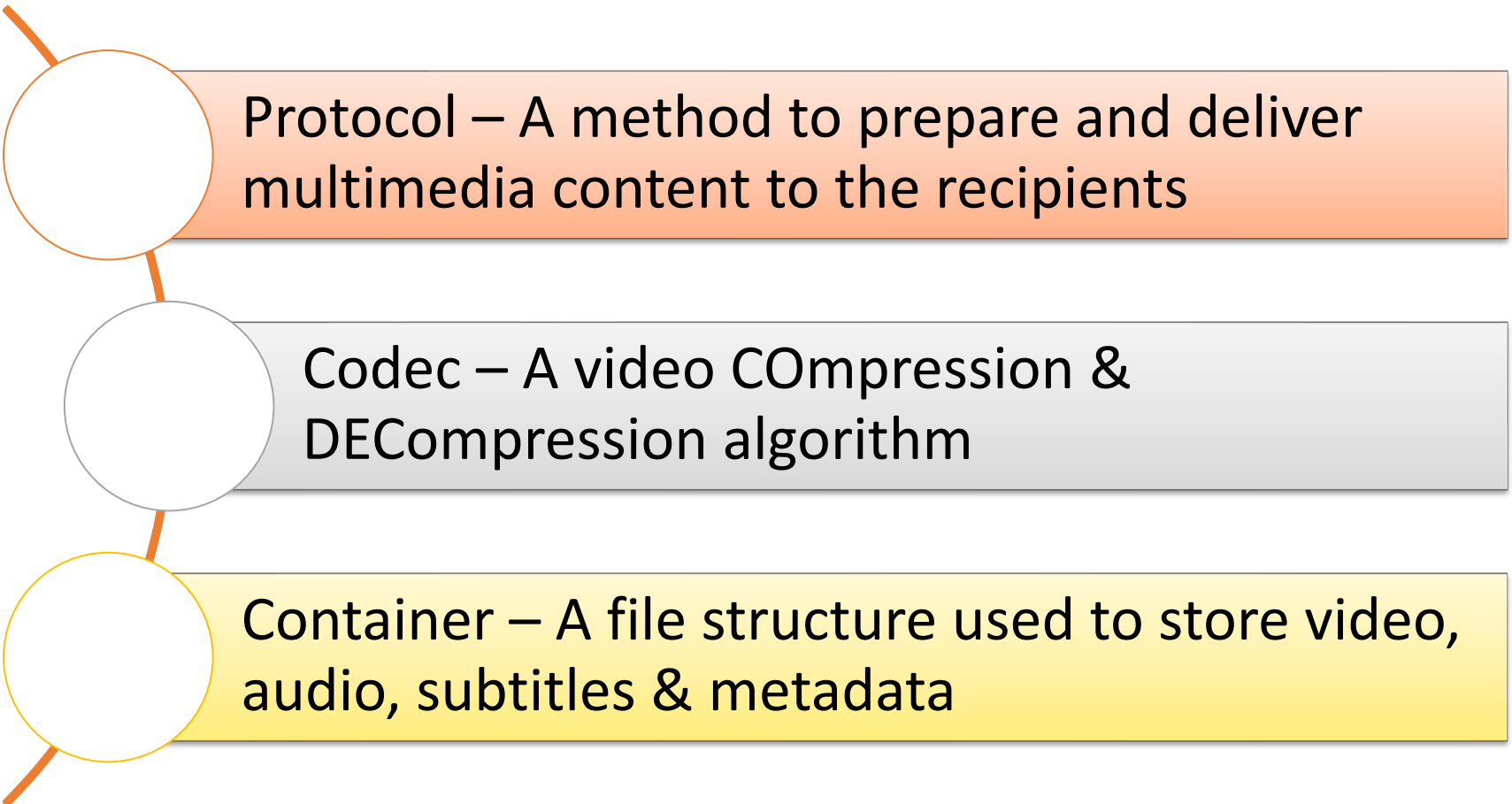
Related videos:

- 3 Hour 4K Wonderful Aerial Views of the Earth with... PRIMAL EARTH SOUNDS 1.1M views • 7 months ago 3:01:40
- FLYING OVER ITALY (4K UHD) Relaxing Music Along With... Homemade Espresso 823K views • 3 months ago 3:50:32
- FLYING OVER FRANCE (4K UHD) - Relaxing Music Along... Homemade Espresso 234K views • 2 months ago 3:55:08
- Maldives 4K - Relaxing Music Along With Beautiful Nature... Piano Relaxing 383K views • 6 months ago 3:48:34
- Mix - 8K VIDEOS ULTRA HD YouTube
- FLYING OVER JAPAN (4K UHD) - Relaxing Music Along With... Piano Relaxing 827K views • 5 months ago 3:49:44
- TIMELAPSE OF THE FUTURE: A Journey to the End of Time (4K) melodysheep 788K views • 2 months ago

Streaming – Protection via buffering



Streaming – Protocol vs Container vs Codec



Protocol – A method to prepare and deliver multimedia content to the recipients

Codec – A video COmpression & DECompression algorithm

Container – A file structure used to store video, audio, subtitles & metadata

Streaming – Protocols

- HTTP Live Streaming (HLS):
 - Released by Apple 2009 (to drop Flash from iPhone).
 - The most widely used streaming protocol now.
 - Widely supported by browsers, TVs or mobile devices (native or via Media Source Extensions).
 - The safest robust, adaptive solution.
 - Latency issues, limitation (cost) of HLS encoders.

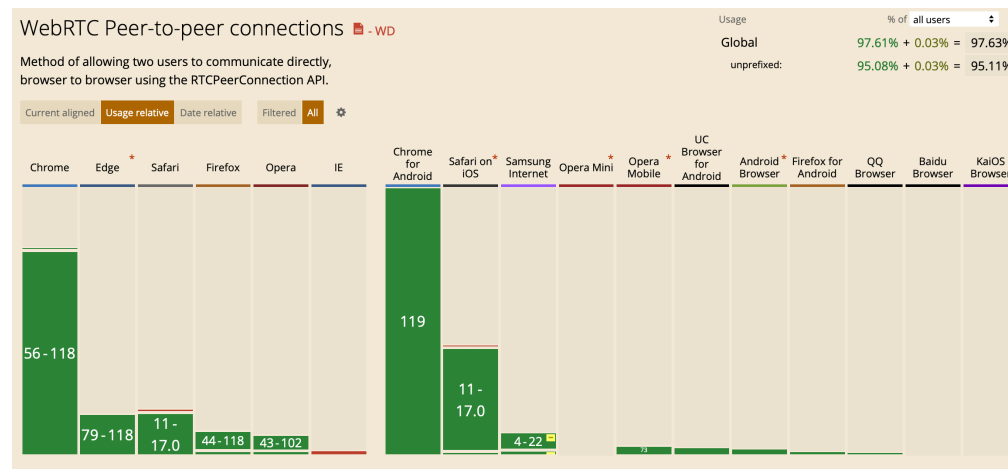
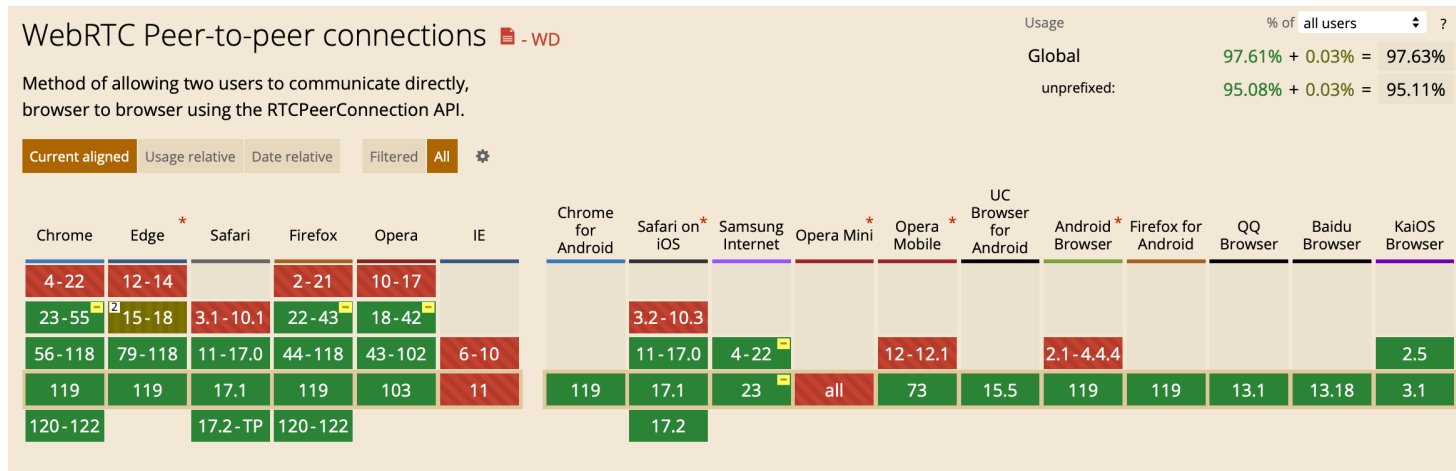
Streaming – Protocols

- Real-Time Messaging Protocol (RTMP):
 - Developed by Macromedia in early days of streaming.
 - Flash dependant, i.e. obsolete protocol.
 - Still used to import live streams by streaming platforms.
 - Not supported by current devices/software used for consumption.

Streaming – Protocols

- Web Real-Time Communications (WebRTC) – mentioned a few slides before:
 - Open source video project
 - Developed to support VoIP.
 - Project purchased by Google.
 - Streaming with real-time latency.
 - Commonly used for web conferencing.

WebRTC support



Streaming – Protocols

- Web Real-Time Communications (WebRTC) – mentioned a few slides before:
 - Open source video project
 - Developed to support VoIP.
 - Project purchased by Google.
 - Streaming with real-time latency.
 - Commonly used for web conferencing.

Streaming - Protocols

- Secure Reliable Transport (SRT):
 - A new solution created by Haivision
 - Supported by SRT Alliance (>500 members) including Microsoft, Alibaba Cloud, Google Cloud, AWS Elemental, Eurovision
 - Security focused
 - Compatibility – device & OS neutral
 - Low latency
 - Not widely supported

Streaming - Protocols

- Real-Time Streaming Protocol (RTSP) – limited support & no streaming over HTTP
- Dynamic Adaptive Streaming over HTTP (MPEG-DASH) - Big potential, but with low level support, lately project stalled
- For a detailed comparison see:
<https://www.dacast.com/blog/streaming-protocols/>

Streaming – Bandwidth calculations

<https://techex.co.uk/streaming-calculator>

- Let's assume we will stream content:
 - bitrate - 2 Mbps
 - length- 90 minutes
 - 1000 watchers
 - Avg. watching time 90 minutes

What is your bit rate? 2 Mbps
Adjust the rate if desired. 2000 kbps

FILE RECORDING

How long will you record? 90 Minutes

Approximate Disk Space needed is:

1350000.00	KB
1318.38	MB (1024 KB)
1.29	GB (1024 KB)
0.00	TB (1024 GB)

Calculate

STREAMING BANDWIDTH

Multicast? ☐

How many viewers will you have? 1000

What is average duration of viewing? 90 minutes

Total viewing minutes: 90000 minutes

Maximum streaming server throughput:

2000000.00	kbps
1953.13	Mbps
1.91	Gbps

Total bandwidth required:

1350000000.00	KB
1318359.38	MB (1024 KB)
1287.46	GB (1024 MB)
1.26	TB (1024 GB)

Calculate

Streaming – Costs (AWS examples)

Cost example 1

Cost example 1 covers a use case of approximately 1,000 viewers viewing a live event for about one hour with a **SD-540p** encoding profile selected in the AWS CloudFormation template. This cost example is based on the following factors:

- Pricing Region: US-East-1, assuming standard pricing (no free-tier or discounts).
- Viewers consume the highest bitrate: Note that bitrate consumption is a mix of all the streams, but the highest bitrate is used in the calculation to show the upper cost range. Additionally, the [Quality-Defined Variable Bitrate](#) (QVBR) and variable video complexity can result in an output bandwidth that is 10-50% lower in price than the estimate provided in Table 1.
- 99% cache/hit ratio between the content delivery network (CDN) and [AWS Elemental MediaPackage](#).

Note

Storage of the test player is not included in this cost estimate.

Table 1 summarizes the total pricing for the live streaming event. Tables 2 through 4 break down the cost for each AWS service.

Table 1: Cost breakdown for 1,000 viewers for a one-hour live event

AWS service	Function	Cost per hour
AWS Elemental MediaLive	Input and outputs for channel	\$1.71
AWS Elemental MediaPackage	Ingest of channel	\$0.14
	Packaging and origination	\$0.40
Amazon CloudFront	Distribution	\$67.24
Total:		\$69.49/hour

Cost example 2

Cost example 2 covers a use case of approximately 10,000 viewers viewing a live event for about one hour with an HD-1080p encoding profile selected in the CloudFormation template. This cost example is based on the following factors:

- Pricing Region: US-East-1, assuming standard pricing (no free-tier or discounts).
- Viewers consume the highest bitrate: Note that bitrate consumption is a mix of all the streams, but the highest bitrate is used in the calculation to show the upper cost range. Additionally, the QVBR and variable video complexity can result in an output bandwidth that is 10-50% lower in price than the estimate provided in Table 6.
- 99% cache/hit ratio between the CDN and AWS Elemental MediaPackage.

Note

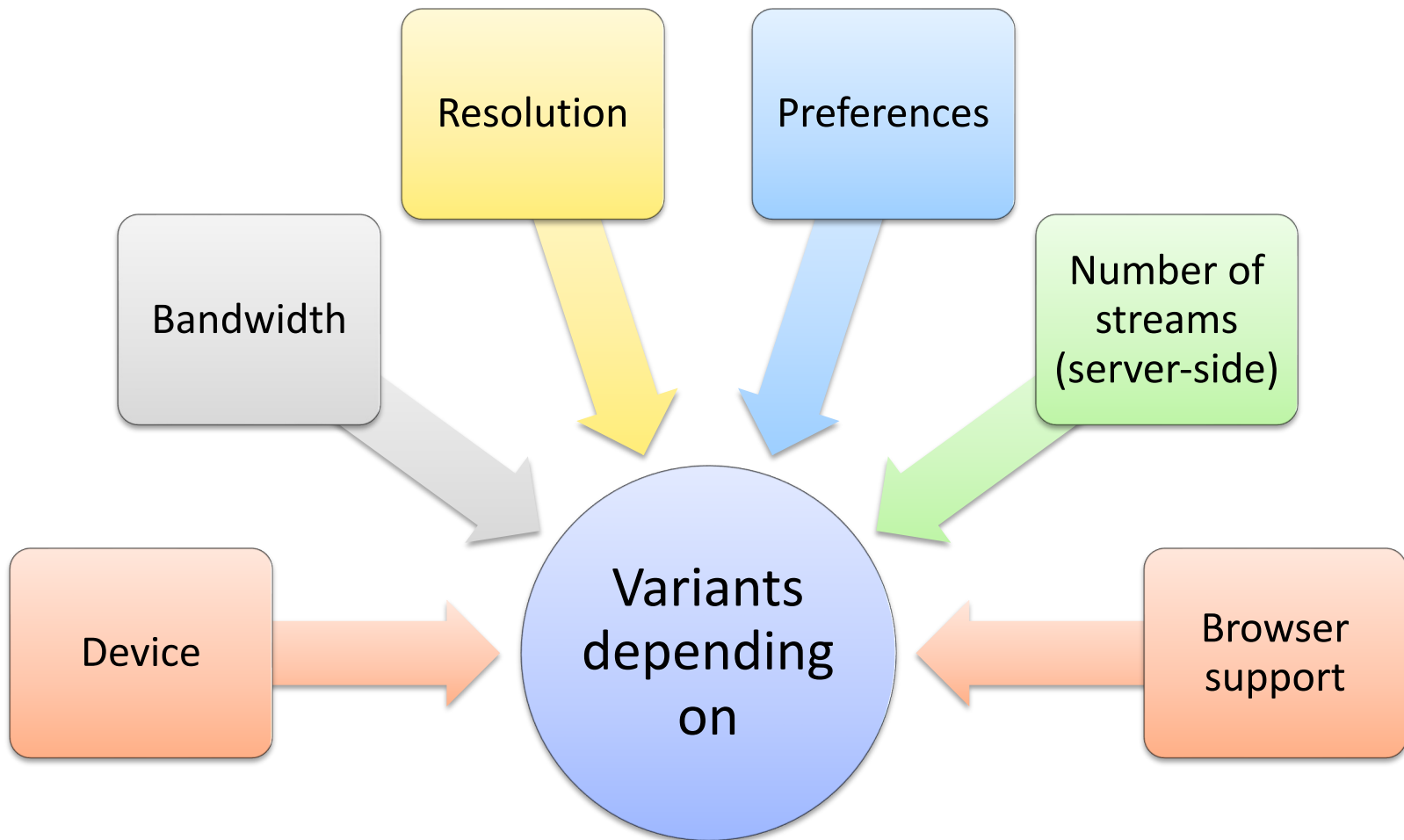
Storage of the test player is not included in this cost estimate.

Table 6 summarizes the total pricing for the live streaming event. Tables 7 through 10 break down the cost for each AWS service.

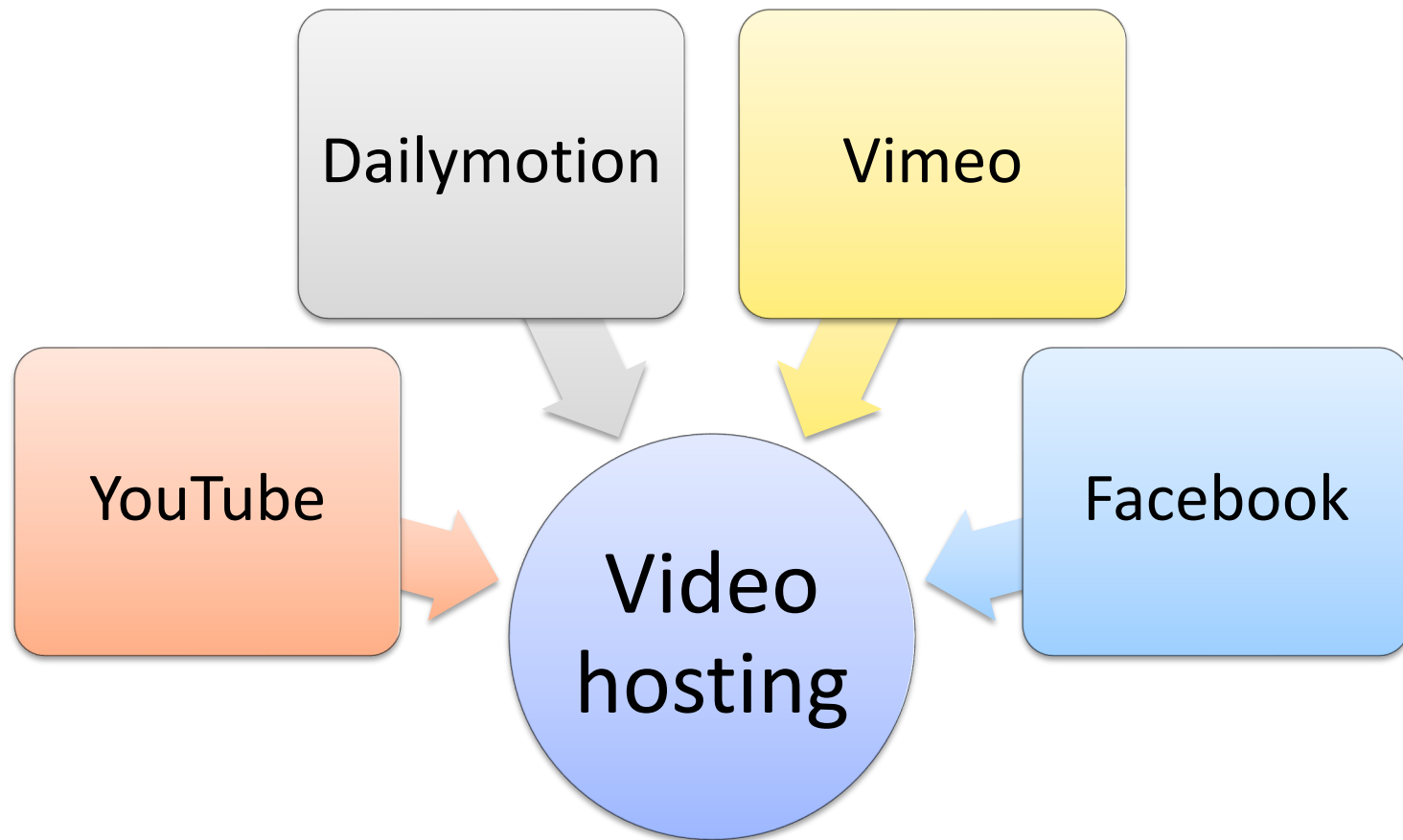
Table 6: Cost breakdown for 10,000 viewers for a one-hour live event

AWS service	Function	Cost per hour
AWS Elemental MediaLive	Input and outputs for channel	\$3.11
AWS Elemental MediaPackage	Ingest of channel	\$0.38
	Packaging and origination	\$9.01
Amazon CloudFront	Distribution	\$1,531.49
Total:		\$1,543.99/hour

Different variants of the same content

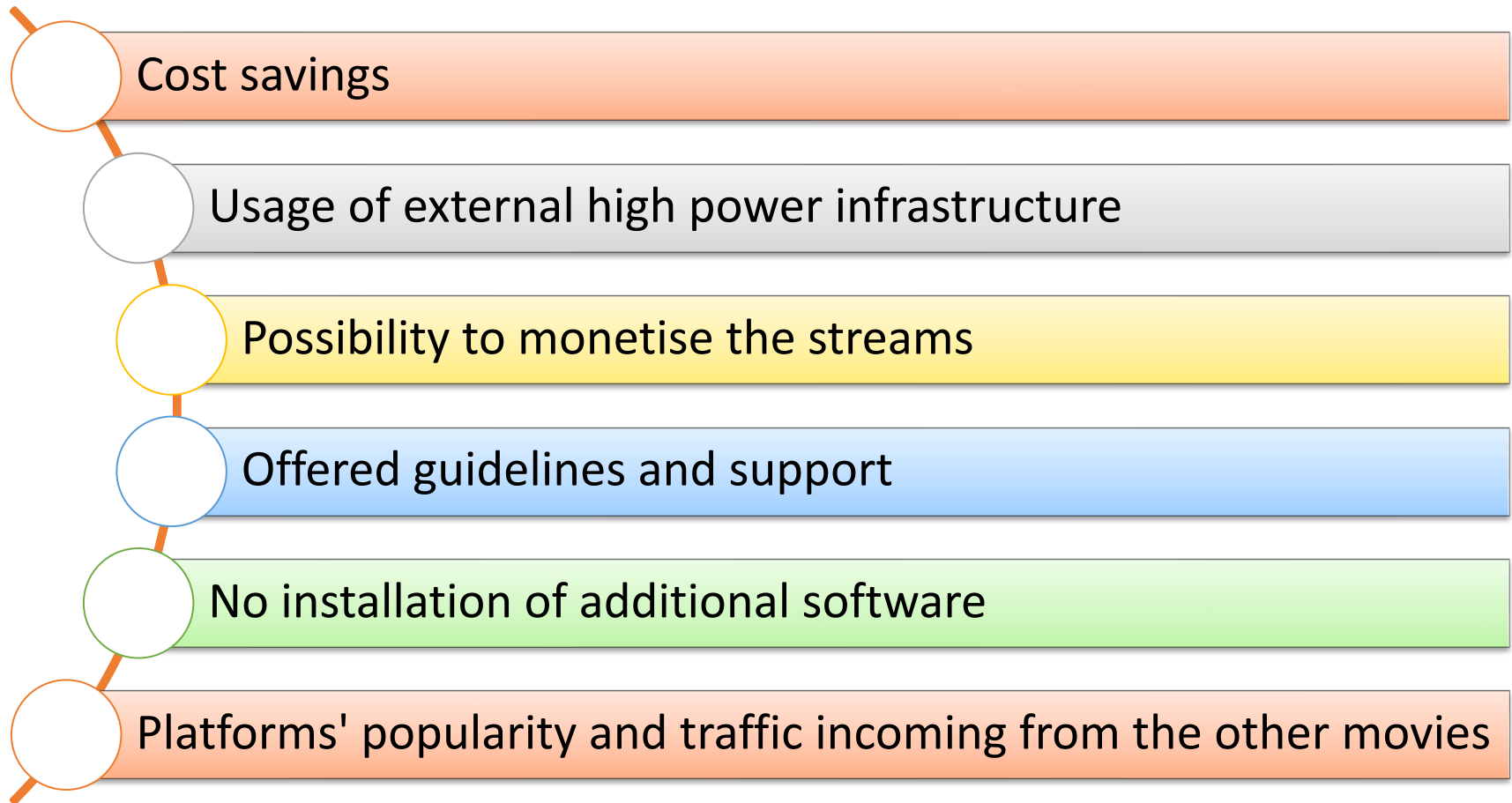


Video (streaming) hosting platforms - free



Note: By embedding the content it can be accessible via company website + the platform.

Reasons to use a streaming platform



YouTube – video requirements

YouTube recommended upload encoding settings



These features are only available to partners who use YouTube Studio Content Manager.

Below are recommended upload encoding settings for your videos on YouTube.

Container: MP4



- No Edit Lists (or the video might not get processed correctly)
- moov atom at the front of the file (Fast Start)

Audio codec: AAC-LC



- Channels: Stereo or Stereo + 5.1
- Sample rate 96khz or 48khz

YouTube – video requirements

Video codec: H.264



- Progressive scan (no interlacing)
- High Profile
- 2 consecutive B frames
- Closed GOP. GOP of half the frame rate.
- CABAC
- Variable bitrate. No bitrate limit is required, though we offer recommended bit rates below for reference
- Chroma subsampling: 4:2:0

Frame rate



Content should be encoded and uploaded in the same frame rate it was recorded.

Common frame rates include: 24, 25, 30, 48, 50, 60 frames per second (other frame rates are also acceptable).

Interlaced content should be deinterlaced before uploading. For example, 1080i60 content should be deinterlaced to 1080p30. 60 interlaced fields per second should be deinterlaced to 30 progressive frames per second.

YouTube – video requirements

Bitrate



The bitrates below are recommendations for uploads. Audio playback bitrate is not related to video resolution.

Recommended video bitrates for SDR uploads

To view new 4K uploads in 4K, use a browser or device that supports VP9.

Type	Video Bitrate, Standard Frame Rate (24, 25, 30)	Video Bitrate, High Frame Rate (48, 50, 60)
8K	80 - 160 Mbps	120 to 240 Mbps
2160p (4K)	35–45 Mbps	53–68 Mbps
1440p (2K)	16 Mbps	24 Mbps
1080p	8 Mbps	12 Mbps
720p	5 Mbps	7.5 Mbps
480p	2.5 Mbps	4 Mbps
360p	1 Mbps	1.5 Mbps

YouTube – video requirements

Recommended video bitrates for HDR uploads

Type	Video Bitrate, Standard Frame Rate (24, 25, 30)	Video Bitrate, High Frame Rate (48, 50, 60)
8K	100 - 200 Mbps	150 to 300 Mbps
2160p (4K)	44–56 Mbps	66–85 Mbps
1440p (2K)	20 Mbps	30 Mbps
1080p	10 Mbps	15 Mbps
720p	6.5 Mbps	9.5 Mbps
480p	Not supported	Not supported
360p	Not supported	Not supported

Recommended audio bitrates for uploads

Type	Audio Bitrate
Mono	128 kbps
Stereo	384 kbps
5.1	512 kbps

YouTube – video requirements

Video resolution and aspect ratio



The standard aspect ratio for YouTube on a computer is 16:9. When uploading other aspect ratios such as vertical or square, the player automatically adapts itself to the size of the video. This setting gives the best viewing experience based on the aspect ratio and device.

Learn how to [use video resolution and aspect ratios](#) correctly.

Recommended resolution & aspect ratios

For the default 16:9 aspect ratio, encode at these resolutions:

- **7680p (8K):** 7680x4320
- **2160p (4K):** 3840x2160
- **1440p:** 2560x1440
- **1080p:** 1920x1080
- **720p:** 1280x720
- **480p:** 854x480
- **360p:** 640x360
- **240p:** 426x240

YouTube – video requirements

Color Space

Recommended color space for SDR uploads

YouTube recommends BT.709 as the standard color space for SDR uploads:

Color Space	Color Transfer Characteristics (TRC)	Color Primaries	Color Matrix Coefficients
BT.709	BT.709 (H.273 value: 1)	BT.709 (H.273 value 1)	BT.709 (H.273 value 1)

YouTube standardizes functionally similar color matrices and primaries before processing the video. For example, BT.601 and BT.709 TRC are identical, and YouTube unifies them to BT.709. Or, BT.601 NTSC and PAL have functionally similar color matrices and YouTube unifies them to BT.601 NTSC. In addition, YouTube may take the following actions to interpret the color space values:

When	YouTube Action
The upload color space has unspecified TRC.	Assumes BT.709 TRC.
The upload color space has unknown or unspecified color matrix and primaries.	Assumes BT.709 color matrix and primaries.
The upload color space mixes BT.601 and BT.709 color primaries and matrix with specified values.	Uses the color matrix to override the color primaries and make them consistent.
The upload color space mixes BT.601 and BT.709 color primaries and matrix, and either primaries or matrix is unspecified.	Uses the specified value of color primaries/matrix to set and override the unspecified one.

Live streaming – Use cases

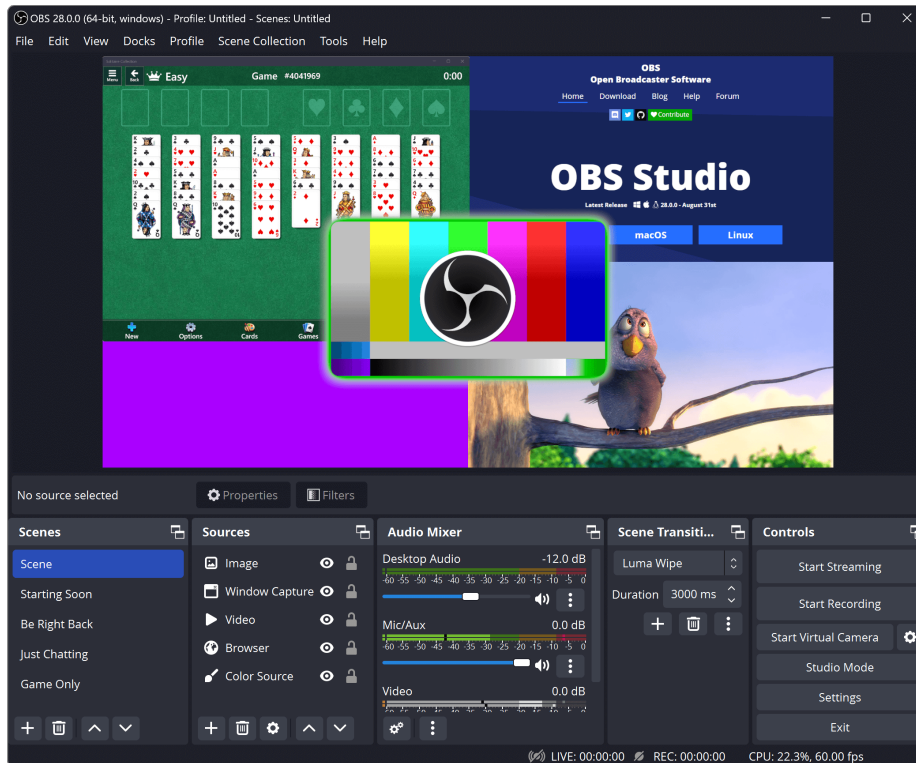


Live streaming - Equipment

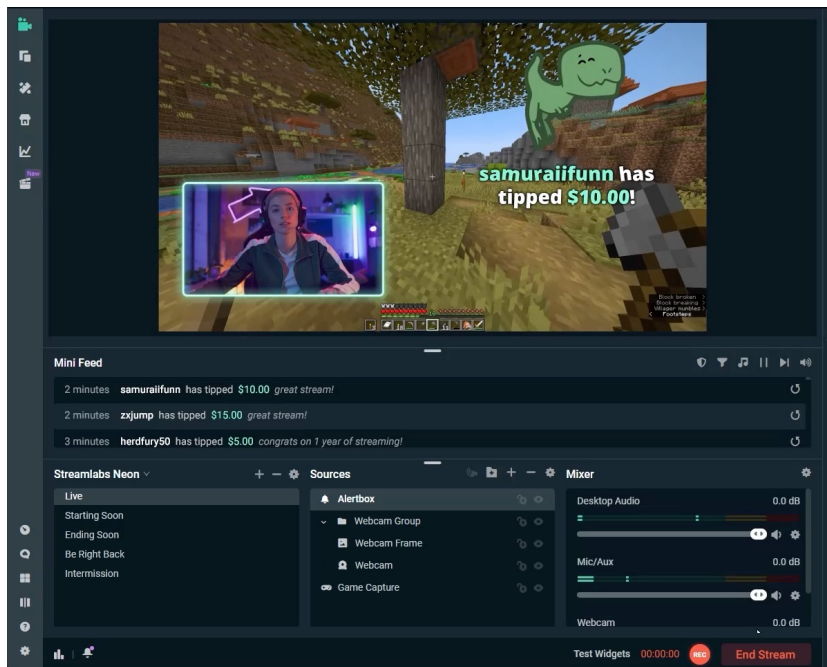


Live streaming – Software #1

- Open Broadcaster Software (OBS Studio)
- Software to catch and mix streams in real time
- Sources:
 - Cameras
 - Multimedia files
 - App windows
- Broadcasting to platforms or acting as a virtual cam



Live streaming – Software #2



- Stream Labs Desktop
- Improved version of OBS
- Build on top of Electron
- Open source project lead by Logitech
- Easy customisation
- Easy to operate

WEB MEDIA – ADDITIONALS

VR / 360° Movies

- 360° movie is taken by several lenses pointed in different directions
- Set of images is transformed into a seamless sphere by so called stitching.



2 fisheye lenses



Stitched image

VR / 360° Movies



VR / 360° Movies

- Still, it is a novelty for web-published movies
- Requires special cameras (more popular)
- It is necessary to perform video processing
- Effect of immersion, especially when camera follows a path and viewers would like to look around, or focus on a part of scene is not feasible
- Integration with VR-devices / rotation via mouse
- See: <https://vr.youtube.com/create/360/>

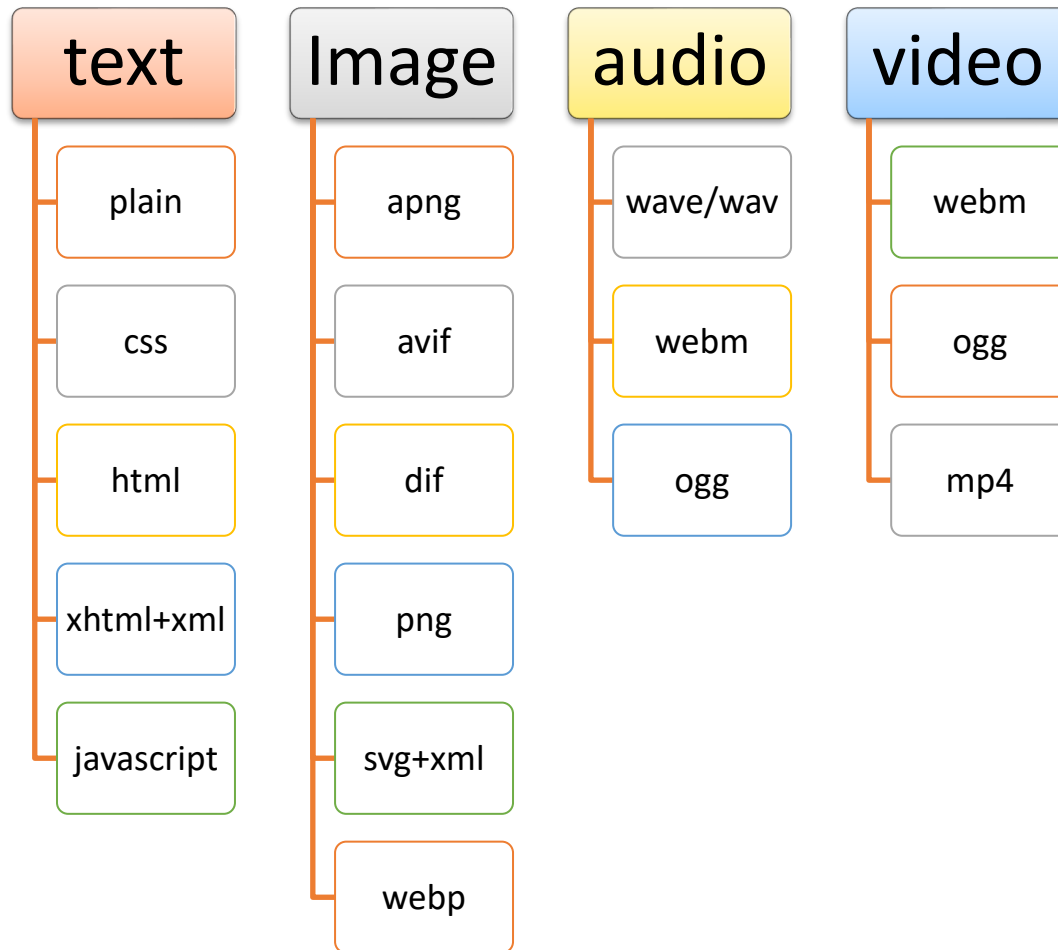
360° Images via Google Street View -> Google Maps



Multipurpose Internet Mail Extensions - MIME Types

- MIME types are commonly (beyond mail) used file type identifiers
- Have a form of `type/subtype`
- Used to describe the type of delivered content i.e. allowing the software to process it
- See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Basics_of_HTTP/MIME_types
- Standardised by the Internet Assigned Number Authority (IANA)

MIME Types – For Web-devs



Summary

- Web = Code + Content (including media)
- Efficient usage of multimedia on Web sites requires selection of proper standards & their parameters
- Not everything is supported by all browsers – cross-check support before making a decision
- In some scenarios the content should be in different forms
- To save the cost – consider a streaming platform.

TILL NEXT LECTURE